

Methods in determining the spatial size of a light source using aperture masking interferometry

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ABSTRACT

Interferometry can be used to measure the spatial size of an astronomical object; however, there are many ways to obtain the interferometric visibility needed to calculate this. Aperture masking was used to generate diffraction patterns for a 2 mm diameter uniformly lit source at different baselines. Two methods were used to calculate the visibility: (1) finding the best-fit parameters to a model of the interference pattern, and (2) using the Fourier Transform of the background-corrected interference pattern. The first method led to a disagreement in the measured diameter by over 3σ , which was likely due to improper calibration of the data and hinted at issues in the model chosen. The second method led to a significant agreement in the measured diameter by within 1σ , highlighting the success in the method. However, it was limited in accurately measuring the visibilities when more than one baseline was used simultaneously. Despite this, it still demonstrated that the method could provide measurements in multiple baselines simultaneously, which offers a practical advantage over the first method in astrophysical interferometry.

1 INTRODUCTION

The ability to resolve fine details of celestial objects is fundamental to observational astronomy. However, because the resolving power of an individual telescope is directly proportional to its aperture size, constructing an instrument with a sufficiently large aperture to resolve surface features of distant celestial objects is a challenging and impractical endeavour (Lawson 2000). Interferometry overcomes this diffraction-limited resolution by using the interference of light, effectively increasing the system’s resolving power. The first formal application of interferometry in astronomy was by Michelson (1891) to measure the sizes of Jupiter’s moons, and was later used to determine the diameter of α -Orionis (Betelgeuse) (Michelson 1920). Since then, astronomical interferometers have become increasingly prevalent.

The principles guiding interferometry techniques are analogous to those governing Young’s double slit experiment, in which the two slits that have a baseline (separation distance) of b act as independent, coherent sources of light waves. The resulting interference pattern arises due to constructive and destructive interference, which depends on the phase differences between the waves as they propagate along different optical path lengths (Fig. 1). More specifically, waves travelling along path lengths that differ by an integer number of wavelengths interfere constructively, whereas those travelling along paths differing by half a wavelength interfere destructively (Monnier 2003; Hariharan 2007). The intensity distribution appears as a pattern of bright and dark fringes separated by λ/b rad, and the complex amplitude A of these fringes is the sum of the complex amplitudes of the individual waves (Hariharan 2007). As the intensity I is directly proportional to $|A|^2$, a more useful parameter to quantify the apparent contrast of the fringes is the Michelson fringe visibility V . This is defined as (Michelson 1920; Michelson & Pease 1921)

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \quad (1)$$

where $I_{\max}$ and $I_{\min}$ are the maximum and minimum intensity of the interference pattern respectively. This normalises the visibility to $0 \leq V \leq 1$. Unlike the double-slit experiment, which relies on a coherent point source, most astronomical sources are extended and incoherent. These sources can be modelled as ensembles of inde-

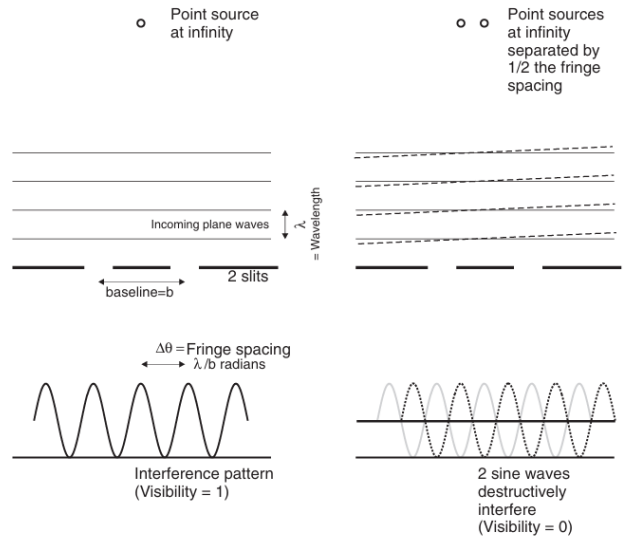


Figure 1. Interference patterns from Young’s double-slit experiment illustrating the fundamental principles of stellar interferometry. **Left:** Intensity distribution from a single point source. **Right:** Intensity distribution from a binary source with a separation of half the fringe spacing, resulting in complete destructive interference and a visibility of $V = 0$. Adapted from Monnier (2003).

pendent point sources, each producing its own fringe pattern. The superposition of these patterns reduces fringe visibility due to partial coherence effects ((b) and (d) in Fig. 2) (Hariharan 2007). Two incoherent point sources separated by $\lambda/2b$ will be completely out of phase, causing their interference fringes to cancel out and produce a uniform intensity distribution. This is analogous to the Rayleigh criterion for the diffraction-limited telescope and defines the resolution limit of an interferometer (Monnier 2003; Hariharan 2007). For smaller angular separations, fringes do not fully cancel but exhibit reduced visibility. Since fringe visibility encodes spatial information about the source, it can be analysed by measuring visibility across a range of wavelengths and baselines. However, a finite wavelength

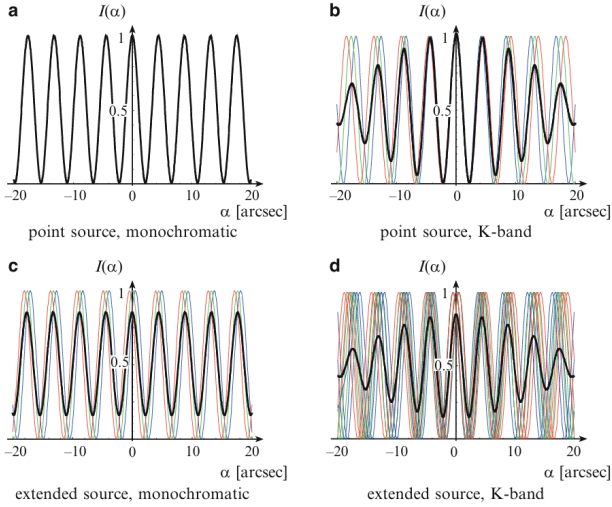


Figure 2. Effect of an extended source and finite wavelength range on the interference pattern with a baseline of $b = 10$ cm. (a) shows the intensity distribution for a monochromatic point source, (b) shows the reduction in visibility due to spectral bandwidth, (c) shows the reduction in visibility caused by an extended monochromatic source, and (d) shows the combined effect of an extended source and spectral bandwidth. Adapted from Glindemann (2011, pg. 59).

range also reduces fringe visibility ((c) in Fig. 2) (Glindemann 2011). Using band filters to restrict wavelengths mitigates this effect.

Astronomers utilise the van Cittert-Zernike theorem to state that at sufficiently large distances, the Fourier Transform of an incoherent source corresponds to its complex visibility. The theorem establishes the relation between an incoherent source and the Fraunhofer diffraction pattern from an aperture of identical shape, illuminated by a spatially coherent wavefront. It provides a proportional relationship between fringe visibility and the amplitude of an image’s Fourier component corresponding to the fringe spatial frequency b/λ . A derivation of the theorem and its assumptions is covered in Thompson et al. (2017, Chapter 15). For a uniformly illuminated circular aperture with an angular diameter θ , the theorem can be used to find that the visibility follows the equation (Glindemann 2011, pg. 55)

$$V_{\text{uniform}} = \left| \frac{2J_1(\pi\theta b/\lambda)}{\pi\theta b/\lambda} \right|, \quad (2)$$

where J_1 is a first order Bessel function and λ is the observed wavelength. This is what is used to model stars.

A variety of astronomical interferometers have been constructed. One technique is “aperture masking”, which involves placing a mask with a series of holes in the optical path of a telescope. These holes create multiple baselines along which light interferes, creating diffraction patterns on the detector. Effectively, the holes act as a small astronomical interferometer, which enables diffraction-limited imaging from telescopes (Monnier 2003). A prominent example of this technique is the Near-Infrared Imager and Slitless Spectrograph (NIRISS) on the James Webb Space Telescope, which utilises a 7-hole non-redundant mask to image the interference pattern from multiple baselines (Doyon et al. 2023). The resulting diffraction pattern can be interpreted using a Fourier Transform, as described by the van Cittert-Zernike theorem. Beyond aperture masking, combining different telescopes to make one astronomical interferometer can go beyond the diffraction limit of an individual telescope. The physical separation between the telescopes serve as the baselines, allowing for angular resolutions far greater than those of any individual

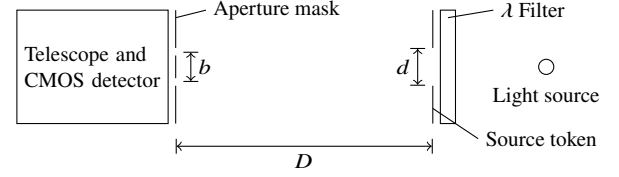


Figure 3. Diagram of the experiment setup used in this study. The distance between the baseline and uniform source was measured to be $D = 8.565 \pm 0.002$ m. The spatial diameter of the uniform source was set as $d = 2$ mm.

optical telescope. For instance, the Center for High Angular Resolution Astronomy (CHARA) array has been used to image β - Cassiopeiae and α - Leonis (Che et al. 2011), while the Event Horizon Telescope (EHT), leveraging data from Very Long Baseline Interferometry (VLBI) stations, successfully captured the first image of the M87 black hole (Event Horizon Telescope Collaboration 2019). Additionally, the MeerKAT radio interferometer has been instrumental in observing the ASKAP J1935+2148, the slowest-spinning long-period radio transient (Caleb et al. 2024). By achieving angular resolutions several orders of magnitude greater than those of individual telescopes, interferometry has enabled astronomers to investigate the fine structure of distant galaxies, protoplanetary disks, and even the event horizons of black holes.

This study investigates the effect of varying baselines and wavelengths on the visibility of interference patterns using aperture masking. Two methods were used to obtain the visibility values: parameter fitting and Fourier analysis. Parameter fitting involves modelling the observed interference pattern by fitting a theoretical curve, while Fourier analysis utilises the van Cittert-Zernike theorem described above. Each method has distinct advantages and limitations, which are examined in this study. Section 2 details the experimental setup. Section 3 discusses both methods in detail and presents their results. Section 4 analyses these results, comparing the two methods within the context of this study and their broader astrophysical applications. Finally, concluding remarks are presented in Section 5.

2 METHODOLOGY

The diffraction pattern at a given b/λ from a uniform light source of diameter $d = 2$ mm was measured using a telescope. The experimental setup is shown in Fig. 3. The telescope used was a 90 mm Celestron C90 MAK Spotting Scope and was equipped with a QHY 5L-II-M CMOS detector, which has a resolution of 1280×960 pixels. The telescope was fitted with a lens cap with a slot to allow a mask with a baseline to be used, blocking all light that did not get diffracted. The light source was placed a distance D away from the telescope, which was measured to be $D = 8.565 \pm 0.002$ m. The light source was set to have a spatial diameter of $\theta = 2$ mm using a source token with an aperture of the same size. Wavelength filters were placed in front of the source, which restricted the mean wavelength of the incoming light to $\lambda = 455$ nm, $\lambda = 530$ nm and $\lambda = 615$ nm when taking into account the quantum efficiency of the CMOS detector. From this, the visibility of the diffraction pattern was measured by both fitting a first order Bessel function and by using a Fourier transform, detailed in Section 3. A list of all the aperture masks used, their respective baselines, the number of holes in each mask and what method was used to calculate their visibility is presented in Table 1.

Uncertainties in the visibility arise from non-uniform illumination of the source aperture, noisy images and dark current in the CMOS detector. A non-uniform illumination of the source aperture

Table 1. List of the aperture masks used. All masks were observed in the 455 nm, 530 nm and 615 nm filters. The visibility from images taken with Mask 6 could only be determined using the Fourier method due to having three interference patterns from the three baselines, instead of one.

Mask No.	Baselines (mm)	n_{holes}	Visibility Determination	
			Parameter Fitting	Fourier
1	0.6	2	Yes	Yes
2	1	2	Yes	Yes
3	2	2	Yes	Yes
4	3	2	Yes	Yes
5	4	2	Yes	Yes
6	0.5, 1, 1.12	3	No	Yes

reduces the measured θ from interferometry due to incorrect visibility measurements. Checking that an image produced is roughly uniform before using a baseline helped to mitigate this issue. Images with poor signal-to-noise ratios will prevent clear identification of the peaks and troughs of the interference pattern. This especially impacts visibility measurements made using a Fourier transform. Taking images with a long exposure time, but low gain, prevented any artefacts due to poor signal-to-noise ratio. However, this does allow for more dark current to influence the image in the CMOS detector. Using a bright light source allows for a shorter exposure time whilst keeping a good signal-to-noise ratio, preventing the dark current from building up during imaging. A dark frame was also subtracted from the image to account for the dark current.

3 RESULTS

3.1 Parameter fitting method

The visibility for an image was determined by generating and fitting a model of the interference pattern to the image. The model was split into three separate components: the point spread function (PSF), the fringes and the background level. The PSF, I_{PSF} , was modelled as a first-order Bessel function parametrised by the centre of the PSF (x_c, y_c) and a scale factor g . The fringes, I_f , were modelled using the equation (Glindemann 2011, pg. 46)

$$I_f(x, y) = I_{\text{avg}} \times \left(V \cos \left[2\pi \frac{x \sin(-\phi) + y \cos(-\phi)}{\lambda_{\text{fringe}}} - \varphi \right] + 1 \right), \quad (3)$$

where $I_{\text{avg}} = (I_{\text{max}} + I_{\text{min}})/2$ (background corrected), ϕ is the angle from the baseline vector ($\phi = 0$ when the fringe pattern aligns horizontally), λ_{fringe} is the wavelength of the fringe pattern and φ is the phase of the fringe pattern. The background was taken as a constant value I_b . The model was then constructed as $I = I_{\text{PSF}} \times I_f + I_b$. An initial guess of I_{max} , I_{min} , x_c , y_c , g , ϕ , λ_{fringe} , φ and I_b was made based on visual inspection of the image. The initial guess for V was made using Eqn. (1). The parameters were minimised using a Markov chain Monte Carlo algorithm, along with minimising the sum of the squared residuals, χ^2 , between the image and the model. This was repeated until a roughly uniform image in χ^2 was obtained.

The visibility as a function of b/λ was plotted and is shown in Fig. 4. This was done for the first five masks in Table 1. A non-linear least squares fit was used to fit Eqn. (2) to the data. The error for all V was estimated as the standard deviation in the residuals between the fit and the data. 9/15 of the data points are within 1σ of the fit, which demonstrates a good fit to the data. The y-intercept given for the fit is $V = 1.09 \pm 0.03$, which is within 3σ of the expected y-intercept of 1. This indicates that the fit is in reasonable

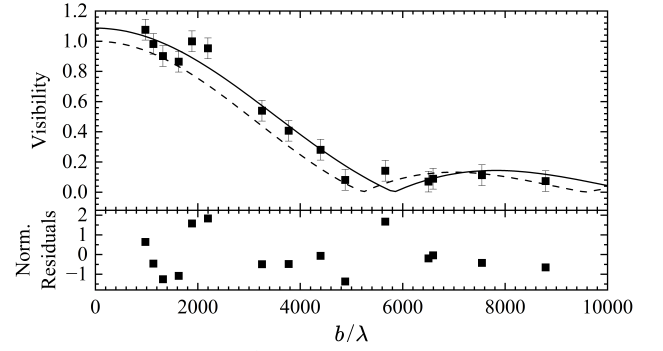


Figure 4. Visibility against b/λ graph for visibilities determined from the parameter fitting method. The dashed line is the theoretical V against b/λ curve modelled by Eqn. (2) for a 2 mm diameter uniform source at the measured D . The solid line is the best fit of Eqn. (2) to the data using a non-linear least squares fit. The fit has a y-intercept of 1.09 ± 0.03 , which is within 3σ of the expected y-intercept of 1. The fit measured the angular diameter of the source as $\theta = (0.209 \pm 0.006) \times 10^{-3}$ rad.

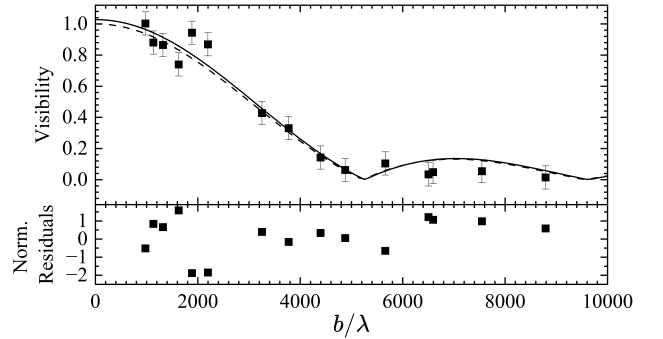


Figure 5. Visibility against b/λ graph for visibilities determined from the Fourier method for Masks 1 to 5. The solid and dashed lines are defined in Fig. 4. The fit has a y-intercept of 1.03 ± 0.04 , which is within 1σ of the expected y-intercept of 1. The fit measured the angular diameter of the source as $\theta = (0.232 \pm 0.009) \times 10^{-3}$ rad.

agreement with the expected curve. However, both the y-intercept and two of the data points are above a visibility of 1, which hint at additional errors in V . The fit resulted in an angular diameter of the source of $\theta = (0.209 \pm 0.006) \times 10^{-3}$ rad. This measured the spatial diameter of the source as $d = 1.79 \pm 0.06$ mm. The uncertainty in d was propagated from the uncertainty in θ and D . The fit determined diameter is more than 3σ away from the actual diameter of the source token and therefore shows disagreement between the values. As the uncertainty in θ is much larger than in D , this further demonstrates errors in measuring V .

3.2 Fourier method

The visibilities in Section 3.1 were redetermined by using the Fourier transform of their respective image and is presented in Fig. 5. The image was corrected for the background level using the best-fit parameters determined, before being Fourier transformed. The Fourier transformed image was then normalised. The visibility was then calculated by reading the value at the secondary peak in the Fourier transformed image and multiplying by n_{holes} for the mask. A non-linear least squares fit was used to fit Eqn. (2) to the data (Fig. 5). The error for all V was estimated as the standard deviation in the residuals between the fit and the data. 10/15 of the data points are within 1σ of the fit, which demonstrates a good fit to the data. The

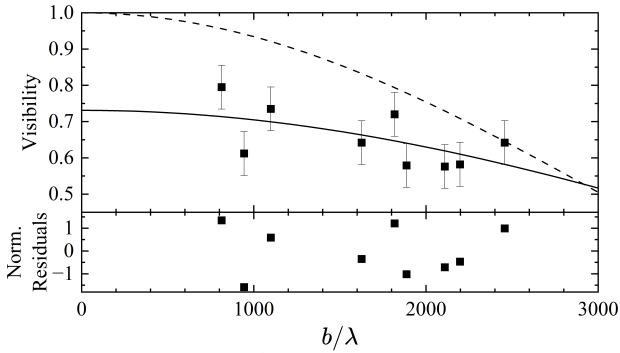


Figure 6. Visibility against b/λ graph for visibilities determined from the Fourier method for Mask 6. The solid and dashed lines are defined in Fig. 4. The fit has a y-intercept of 0.73 ± 0.04 , which is greater than 3σ from the expected y-intercept of 1.

y-intercept given for the fit is $V = 1.03 \pm 0.04$, which is within 1σ of the expected value of 1 and indicates significant agreement with the predicted curve. The angular diameter of the source measured by the fit was $\theta = (0.232 \pm 0.009) \times 10^{-3}$ rad. This gave the spatial diameter of the source as $d = 1.99 \pm 0.07$ mm, which is within 1σ of the expected value. The uncertainty in d was propagated from the uncertainty in θ and D . This shows significant agreement between the Fourier determined diameter and actual diameter.

Additionally, the visibilities for the last mask in Table 1 was determined using the Fourier transform of the image and is presented in Fig. 6. The mask contained three baselines, which created three superimposed interference patterns. A model that recreates this is hard to implement, so this image was omitted from the previous method. I_b was set as the median intensity of the image. The image was then corrected for I_b and Fourier transformed, where the secondary peak in the Fourier transformed image for each baseline was identified by visual inspection. The visibility for each b/λ was then the value at the secondary peak multiplied by n_{holes} . A non-linear least squares fit was used to fit Eqn. (2) to the data (Fig. 6). The error in V was estimated as the standard deviation in the residuals. The y-intercept given for the fit is $V = 0.73 \pm 0.04$, which is more than 3σ away from the expected y-intercept of 1. This shows that the visibilities for the mask disagree with Eqn. (2), suggesting underlying errors with the measurements.

4 DISCUSSION

The parameter fitting method produced a fit with a y-intercept of 1.09 ± 0.03 that resulted in a measured source diameter of $d = 1.79 \pm 0.06$ mm. The Fourier method produced a fit with a y-intercept of 1.03 ± 0.04 that resulted in a measured source diameter of 1.99 ± 0.07 mm. The y-intercept for the fitting method is within 3σ of the expected value of 1. However, the measured source diameter is more than 3σ away from the actual diameter of $d = 2$ mm showing the values are not in agreement. For the Fourier method, both values agree with their expected values as they are within 1σ . Using this method for the last mask in Table 1 resulted in a y-intercept of 0.73 ± 0.04 which is greater than 3σ away from the expected value.

The visibilities in the 2, 3, and 4 mm baselines were consistent with each other in that the values calculated for different source wavelengths at the same baseline followed the same trend. This is in contrast to the values at the 0.6 and 1 mm baselines which appear more sporadic around the fit. A possible reason for the inconsistency in values with a baseline of 0.6 mm is that the interference patterns did

not completely fit onto the CCD. These smaller baselines correspond to higher visibilities so the higher intensity of these images could be a cause. Another improvement to the data could be to increase the difference between the interference pattern and the background by using a darker room, although the difference in this experiment was large enough that this may not have noticeably affected the results. A more significant trend in the data can be seen in Fig. 4 which appears to have a systematic error when comparing the line of best fit to the theoretical line and results in the visibility straying out of its constraints of $0 \leq V \leq 1$. This would likely have been solved by a zero-point calibration on the visibility by observing a point-like source. This error is less obvious in Fig. 5 but would likely still have affected the line of best fit. Both fits would also have improved accuracy and precision if more data points were taken.

Out of the two methods, the Fourier method produced a diameter measurement much more accurate to the actual value, although the fitting method was slightly more precise. The smaller baseline data points surpassing a visibility of 1 in Fig. 4 suggests that the fitting method struggles at higher intensities and is more significantly influenced by the zero-point calibration. This could also possibly be true for visibilities close to zero as the data point closest to the minimum of the line of best fit is noticeably higher than this line. The fringes of the interference pattern get closer together as the visibility gets closer to zero meaning that it could become harder to use this method accurately for longer baselines. Despite that, this was not a difficulty that was encountered during this experiment. The normalised residuals of the fitting method visibilities look to be scattered overall but there may be some pattern within each baseline. Using the Fourier method on Mask 6 produced a value for the source diameter that was very different from the actual diameter. An error this large was likely caused by a fault with the mask leading the interference pattern to be incorrect. However, calculating the diameter this way shows that using the Fourier method allows multiple baselines to be used at once. Along with the more accurate results, this would suggest that the Fourier method is the more practical approach for use in astronomical interferometry as data could be collected more efficiently.

5 CONCLUSION

Interferometry is a revolutionary technique that overcomes the diffraction limit of an individual telescope by combining light signals from multiple telescopes separated by large distances. This effectively increases the system's resolving power by simulating a much larger aperture. This study investigated the effects of varying baseline lengths and wavelengths on the visibility of interference patterns using the technique of "aperture masking" and estimated the spatial size d of an extended light source. Two methods were used to determine the visibility required for calculating d : parameter fitting and Fourier analysis. It was found that Fourier analysis produced a more accurate estimate of 1.99 ± 0.07 mm, which agrees with the actual value of 2 mm within 1σ . In contrast, the parameter fitting method yielded a value of $d = 1.79 \pm 0.06$ mm, which deviates by 3σ from the true value. Despite the Fourier analysis being limited in accurately measuring visibility using the three-hole aperture mask (Mask 6), it proved to be a robust method for measuring visibilities from multiple baselines simultaneously - an approach impractical for parameter fitting. The experiment reinforces the reliability of interferometry in obtaining spatial information about light sources, underscoring its significance in observational astronomy.

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